33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F22-SR-001
415V CARBON TREATMENT & ELUTION SWITCHROOM

F22-TR-030
TRANSFORMER ROOM

F22-TR-030
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
4000/ 1A
2.5VA
5P20

F22-SW-030
CARBON TREATMENT & ELUTION 415V MOTOR CONTROL CENTRE No.1

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
Cl 1
4000/ 5A
30VA
Cl0.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
4000A
2A
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6

415V 3NE~ 50Hz 4000A 65kA for 1sec

INC

1F-01	2F-11	3F-11	1F-13	4R-01	3R-11	2R-11	4R-13	4F-05	2F-08	2R-05	3R-01
630A	630A	630A	630A	630A	630A	630A	630A	250A	63A	100A	160A
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
250A	250A	250A	250A	250A	250A	250A	250A	90A	70A	100/1A	100/1A
	EOL	EOL			EOL	EOL		EOL			
3PX			3PX	3PX			3PX				
168kW	168kW	168kW	168kW	168kW	168kW	168kW	168kW	75kW	22.5kW	30kW	45kW
F22-KN-411A	F22-KN-411B	F22-KN-411C	F22-KN-411D	F22-KN-411E	F22-KN-411F	F22-KN-411G	F22-KN-411H	F22-PP-460	F22-HT-411	F22-PP-490	F22-PW-441
CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 1 2000-F22-SCM-EL-10207	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 2 2000-F22-SCM-EL-10208	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 3 2000-F22-SCM-EL-10209	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 4 2000-F22-SCM-EL-10210	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 5 2000-F22-SCM-EL-10211	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 6 2000-F22-SCM-EL-10212	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 7 2000-F22-SCM-EL-10213	CIL4 CARBON REGENERATION KILN HEATERS ZONE 1 BANK 8 2000-F22-SCM-EL-10214	CIL4 ELUTION HEATER THERMAL OIL PUMP 2000-F22-SCM-EL-10223	CIL4 ELUTION HEATER 2000-F22-SCM-EL-10225	CIL4 BARREN CARBON PUMP 2000-F22-SCM-EL-10224	CIL4 ELUTION PUMP 1 2000-F22-SCM-EL-10227

E-SHUT
F22SR001-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F22-SLD-EL-10202

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F22-SW-030 INCOMER TIER." CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF THE SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F22-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET
- 3PX — THYRO POWER CONTROLLER THYRO-PX-3PX-500-350-H-F WITH TOUCH DISPLAY OPERATED IN TAKT MODE (ZERO CROSS FIRING)

Approved	☒
Approved as Noted	☐
Not Approved	☐
For Information Only	☐

Signed [redacted] Date 06/01/2026

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F22-SLD-EL-10201_Rev1

CIL4 carbon treatment & elution F22-SW-030 415V MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELN\F22 - CIL4 CARBON TREAMENT & ELUTION\2000-F22-SLD-EL-10201.DWG 18DEC25 12:14PM HOTSEAT04